Quantum Mechanics II Excercise 2

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1 The wave functional of the vacuum

1.1 The first oder ODE for the ground state of a harmonic oscillator

The ground state of the quantum harmonic oscillator is defined as the state that is annihilated by the ladder operator $\hat{a}$:

$$\hat{a}\,|0\rangle = 0 \tag{1}$$

$\hat{a}$ can be expressed in terms of $\hat{x}$ and $\hat{p}$ we get:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right) \tag{2}$$

Assigning this to equation 1, we get:

$$\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)|0\rangle = 0 \tag{3}$$

In the position representation, $\hat{x} = x$ and $\hat{p} = -i\frac{\partial}{\partial x}$, which leaves us with the first order ODE:

$$\left(x + \frac{\hbar}{m\omega}\frac{d}{dx}\right)|0\rangle = 0 \tag{4}$$

Solving this gives the gaussian:

$$|0\rangle = \exp\left[-\frac{m\omega x^2}{2\hbar}\right] \tag{5}$$

1.2 The wave functional of the ground state of the scalar field

Using the formulation we did in class, we can write the ladder operators for the scalar field: